Energy-Aware Adaptive Large Neighborhood Search for Multi-UAV Coverage Path Planning

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***Abstract*—Multi-UAV coverage path planning assigns sweep routes to a fleet so that the sensor footprint passes over an entire region of interest. Route length is a weak proxy for battery drain, because turns, accelerations, and braking can significantly affect the energy consumption of a multirotor. A recent energy-aware method decomposes the region with Boustrophedon Cellular Decomposition, generates back-and-forth patterns for each cell and casts the assignment as a Multiple Set Traveling Salesman Problem, which it solves with GRASP and Tabu Search. We replace that solver with Energy-ALNS, an Adaptive Large Neighborhood Search whose destroy, repair and local search operators score every decision by estimated energy. Energy-ALNS reuses the decomposition, the energy model and the sweep-angle search of the reference method, allowing a controlled comparison in which only the optimization solver is replaced. Over 16 scenarios and five random seeds, Energy-ALNS cuts the energy makespan of the original solver by 2.9–8.6% in near-convex geometries and matches it, within seed variability, elsewhere. An operator study attributes that gain to the energy-aware operators rather than to a reinforcement learning controller. A Gazebo Sim 8, ROS 2 Humble and PX4 run confirmed that the exported routes execute under multirotor dynamics.**

***Index Terms*—Coverage Path Planning, Multi-UAV Systems, Adaptive Large Neighborhood Search, Energy-Aware Mission, Metaheuristics**

I. Introduction

Coverage Path Planning (CPP) looks for a set of paths that drives the sensor footprint of one or more platforms over an entire region of interest while respecting geometric and operational constraints. Unmanned Aerial Vehicles (UAVs) apply CPP in infrastructure inspection, mapping, environmental monitoring, remote sensing, and post-disaster assessment [1]. Fleets extend the reach of these missions, and swarm research has moved from single-vehicle autonomy toward coordinated operation over large areas [2].

The multi-UAV case (mCPP) demands more than a sweep trajectory. A planner must split the region into subareas, decide which vehicle flies each part of the mission, pick a coverage pattern for every cell and order the patterns along each route. Traveled distance, mission time, return to base, no-fly zones and the limited endurance of each vehicle all shape that decision [1], [3]. Cooperative planners also have to balance the load, since the slowest vehicle sets the makespan of the whole mission [4], [5].

Minimizing total route length, the usual objective of classical CPP, does not minimize the energy a multirotor spends. Turns, decelerations, and heading changes raise consumption above straight flight, so two plans of equal length can drain batteries at different rates. Battery capacity constrains how long a fleet keeps flying, and studies on recharging and charge scheduling treat endurance as the operational bottleneck of UAV fleets [6], [7]. Recent planners answer this by pushing energy estimates into the planning loop, so that the battery limit shapes the routes while they are still being built [1].

Datsko et al. [1] propose the energy-aware reference method we build on. Their pipeline applies Boustrophedon Cellular Decomposition (BCD), generates several back-and-forth coverage patterns for each subpolygon, estimates the energy of every pattern with a fast model and turns the result into an instance of the Multiple Set Traveling Salesman Problem (MS-TSP). A heuristic combining GRASP with Tabu Search solves that instance. We refer to it as the MSTSP solver.

We keep this formulation and swap the MSTSP solver for Energy-ALNS, a solver we derive from Adaptive Large Neighborhood Search [8] and specialize for battery consumption. Its removal, repair, intra-route local search and pattern orientation operators all rank candidates by estimated energy instead of traveled distance, for a homogeneous fleet.

The main contributions of this work are:

- We propose Energy-ALNS, an energy-aware ALNS solver specifically designed for the MS-TSP formulation arising from BCD-based multi-UAV coverage, incorporating seven removal and two repair operators guided by the mission energy model.
- We introduce an intra-route local search coupled with pattern orientation optimization and a final inter-route energy rebalancing stage, both driven by the same energy cost function used inside the adaptive loop.
- We benchmark the solver over 16 scenarios with five seeds each, measuring energy makespan in Wh against MSTSP, DARP [3] and DR-ALNS [9], and we isolate

the source of the gain by transplanting our operators into DR-ALNS.
- We export the planned routes to a ROS 2 Humble, Gazebo Sim 8 and PX4 stack and verify their execution by three simulated multirotors considering acceleration and braking dynamics.

II. Related Work

Datsko et al. [1] report the work closest to ours. They plan energy-aware multi-UAV coverage missions using optimal flight speed, a fast energy estimator, Boustrophedon Cellular Decomposition, several sweep patterns per cell and an MS-TSP formulation solved by GRASP and Tabu Search. We inherit that structure and change only the solver.

A second family divides the region before routing it. Apostolidis et al. [3] present a cooperative mission planning platform for remote sensing whose DARP algorithm partitions the area from the initial positions of the vehicles and then covers each subregion with a spanning-tree sweep. Field and simulated experiments confirm full coverage, but the partition follows geometry and ignores the energy model, which makes DARP a useful geometric baseline for our comparison. Yang et al. [10] replan the coverage tree locally when map updates free new areas, and report higher coverage rates than conventional STC in warehouse simulations. Mitra and Saha [5] plan concurrently for robots that finish their routes while the others keep flying, and measure a 1.6× speedup with up to 512 robots in unknown environments. Neither method models battery consumption while building the routes.

Coverage control research approaches the same coordination problem from continuous control. Cheng et al. [11] survey the area and identify reinforcement learning and adaptive distributed control as the dominant directions. He and Cheng [12] couple task assignment with path generation, Qu et al. [13] enforce persistent coverage under energy limits with control barrier functions, and Ning et al. [14] optimize coverage and network connectivity together. These works control agents in real time and skip the offline route construction our problem requires.

Ropke and Pisinger [8] introduced ALNS for routing problems with coupled decisions, and Santini et al. [15] measured how the acceptance criterion shifts the balance between intensification and diversification. Zhang and Liu [16] bring energy into the operators themselves and apply the resulting EA-ALNS to heterogeneous multi-robot task allocation. Reijnen et al. [9] take a different path with DR-ALNS, where a reinforcement learning policy selects operators online. We use DR-ALNS as an external reference: our solver runs a classical ALNS loop with no learning component inside the planning cycle.

Earlier work from our group addresses adjacent UAV problems with learning. Oliveira et al. [17] sequence waypoints for environmental monitoring with an attention-based policy network over a Traveling Salesman formulation, and Silva et al. [18] fly a single UAV around a farm perimeter using D4PG. Both handle one vehicle over fixed points and leave out area decomposition and fleet coordination. This paper extends that line to energy-aware multi-UAV coverage with a combinatorial solver that assigns and orders routes across vehicles.

Our work sits between the energy-aware BCD line and the ALNS line. We keep the energy model and the MS-TSP formulation of the reference method and measure what an energy-aware adaptive search adds to the quality of the produced routes.

III. Methodology

We adapt the energy-aware multi-UAV coverage pipeline of Datsko et al. [1] so that an Adaptive Large Neighborhood Search solver can take over the assignment stage. The Boustrophedon decomposition, the coverage pattern generation, the energy model and the sweep angle selection stay compatible with the original MS-TSP instance, which lets Energy-ALNS replace the MSTSP solver without touching the rest of the pipeline. One variable separates the two methods: the solver that selects, assigns and orders the patterns.

A. System Overview

Fig. 1 summarizes the planning flow. The system reads the region of interest, the no-fly zones, the initial positions of the UAVs and the physical parameters of the energy model. It rotates the region over a set of candidate angles, decomposes each rotation into BCD cells, converts the cells into coverage patterns and assembles an MS-TSP instance. Each box in the figure maps to one subsection below. Energy-ALNS replaces the optimization component of the original framework, while the remaining stages are retained.

B. Decomposition and Pattern Generation

Boustrophedon Cellular Decomposition splits the region into subpolygons that a systematic back-and-forth sweep can cover. This stage is retained without structural modifications from [1], since Energy-ALNS operates directly on the resulting MS-TSP instance. Each resulting cell receives a set of back-and-forth patterns, and each pattern carries a sweep orientation, a start point, an end point, a waypoint sequence and an estimated energy cost.

The initial rotation of the region changes the decomposition and therefore the available patterns. The planner derives the candidate angles from the orientations of the polygon edges, ranks them by decomposition cost and keeps the seven best that stay apart by more than a fixed angular tolerance. This criterion comes from the reference method [1] and yields unevenly spaced angles that follow the geometry of each region. The planner builds one MS-TSP instance per rotation, runs the solver on each and keeps the best result.

Every pattern carries its own execution cost, given by the energy needed to fly its trajectory, plus a transition cost from the end of one pattern to the start of the next. The problem becomes an MS-TSP: exactly one pattern per cell enters the solution, and the planner distributes and orders the selected patterns among the UAVs to minimize the cost of the most expensive route [1].

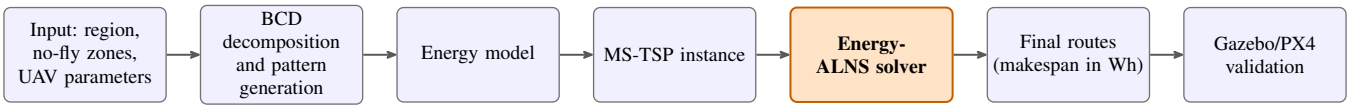


Fig. 1. Planning pipeline adapted from the reference method [1]. The decomposition, the energy model and the MS-TSP construction stay unchanged. The highlighted Energy-ALNS solver replaces the original MSTSP solver, and the Gazebo/PX4 stage checks that the exported routes fly under multirotor dynamics.

C. Energy Model

We estimate consumption with the fast model of Datsko et al. [1], also reused without change because it defines the cost that MSTSP and Energy-ALNS both consume on the same instance. The model estimates energy consumption directly from the waypoint sequence, without requiring the generation of a complete dynamic trajectory, so the planner can score many alternatives during the search. The experiments keep the vehicle configuration of the reference method [1], a 3.2 kg airframe with four propellers of 0.19 m radius, from which the model derives an optimal speed of 8.39 m/s, a hover power of 426.03 W and a straight flight power of 465.23 W. The maximum acceleration of 2 m/s^2 and the seven candidate rotation angles come from the same source, and we tuned no parameter for this study.

Energy-ALNS ranks a solution S with

$$C(S) = E_{max}(S) + w_{sum} E_{total}(S), \tag{1}$$

where E_{max} is the cost of the most expensive route, E_{total} is the sum over all routes and w_{sum} is a small weight that breaks ties toward balanced solutions when two candidates share the same maximum cost.

D. Proposed Energy-ALNS Solver

Energy-ALNS extends the classical ALNS framework [8] by incorporating operators and search mechanisms specifically designed for the energy-aware objective. The name separates it from generic ALNS and from the learning-based DR-ALNS. We implement it in C++ behind the same interface as the original MSTSP solver. A GRASP strategy builds the initial solution: the solver evaluates candidate insertions and draws one from a restricted candidate list. The adaptive loop then explores larger neighborhoods through destroy and repair operators.

Energy-ALNS departs from a conventional ALNS in the criterion behind each decision. Removal, repair and local search all read the cost function $C(S)$ built on the estimated energy model [1]: removal operators target cells that drive E_{max} or overload the heaviest route, and repair operators price every candidate insertion through the same model. Battery consumption guides the solver from the initial construction to the acceptance test.

Fig. 2 details the internal architecture. Seven removal operators feed the loop: random removal, removal of the largest contributors to E_{max}, removal of spatially close clusters, removal of the costliest element of the heaviest route, removal of a contiguous route segment, temporary emptying of the highest-consumption route, and a big shake that drops 35% to 55% of the cells to force stronger restructuring. Two repair operators rebuild the solution. Greedy insertion picks the combination of cell, position and orientation variant with the smallest incremental $C(S)$. Regret-2 insertion prioritizes cells whose second best energy insertion falls far behind the first, which keeps expensive cells from being left to the end of the reconstruction.

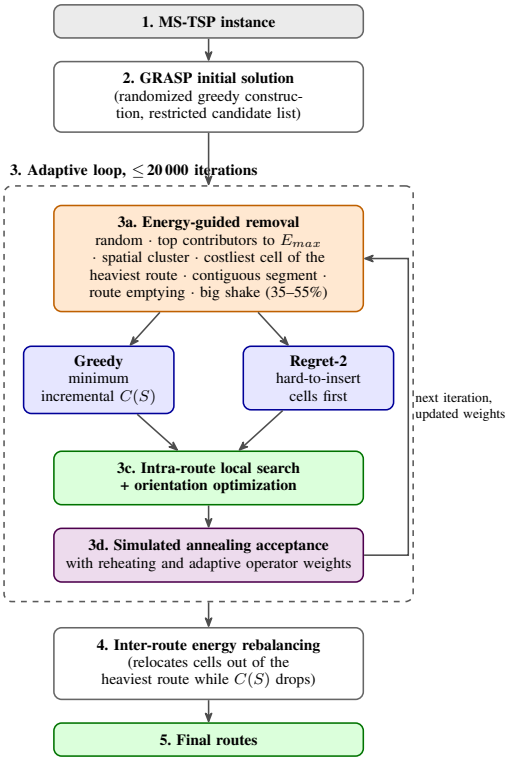


Fig. 2. Internal architecture of the proposed Energy-ALNS solver. Every stage of the loop decides through the energy cost $C(S)$, and no stage ranks candidates by pure distance.

Repair is followed by an intra-route local search combined with pattern orientation optimization. The local search moves short segments inside a route and measures the impact on E_{max}, while the orientation step tries alternative pattern variants for the same cell. A Simulated Annealing criterion then compares the candidate cost against the incumbent: the solver accepts every improvement and accepts worsening moves with a probability that decays along the search. Reheating at regular intervals restores that probability and pulls the search out of local optima. The framework updates the weights of the removal and repair operators from their recorded performance. The loop runs for up to 20 000 iterations with $w_{sum} = 10^{-2}$ and stops early after 7 500 iterations without improvement.

This budget keeps the planner under 13 s in the largest instance (Table I); a sensitivity analysis over it remains open.

Once the loop ends, a final inter-route energy rebalancing stage takes the best solution found. For each cell of the highest-consumption route, the solver tests a relocation to the remaining routes and accepts the first move that lowers $C(S)$, repeating the sweep over a few passes. This stage adjusts the load across routes without going through destroy and repair, and produces the routes we export.

IV. Experiments

The experiments compare Energy-ALNS against three methods: the MSTSP solver of the reference work [1], which applies GRASP and Tabu Search over the same MS-TSP formulation; DARP [3], which partitions the area by a geometric criterion; and DR-ALNS [9], an ALNS whose operator selection follows a reinforcement learning policy.

A. Experimental Settings

We run 16 scenarios derived from the reference paper, split into six near-convex configurations (Simple and Rectangle, each with 2, 3 and 4 UAVs) and ten irregular configurations or configurations with a no-fly zone (Cape, Complex, Complex Large and Island, over different fleet sizes). Each stochastic solver (MSTSP, Energy-ALNS, DR-ALNS) runs with five independent random seeds. DARP is deterministic and runs once per scenario. The primary metric is the energy makespan in Wh, computed by the energy evaluator shared by all solvers and defined as the consumption of the UAV that spends the most during the mission.

Five geometric types cover the 16 scenarios. Simple and Rectangle are near-convex polygons, with at most one reflex vertex and no obstacle. Cape is a non-convex area with deep lateral indentations, which forces BCD into elongated cells and constrains load redistribution across UAVs. Complex and Complex Large are irregular polygons with several concavities, the second reproducing the first at a larger scale. Island holds a central no-fly zone, so the UAVs cover the ring around the obstacle, which cuts the number of viable partitions. Fig. 3 shows routes that Energy-ALNS produced for one representative of each type.

Besides the makespan, we record the traveled distance and the computation time. DARP partitions the area without consulting the energy model, so we keep it as a reference for an alternative coverage paradigm and leave it out of the win count against MSTSP. All runs use a laptop with an Intel Core i7-14650HX, 32 GB of RAM and an RTX 4060 GPU under Ubuntu 22.04 LTS, with planning and benchmark on CPU alone.

B. Energy Makespan per Scenario

Table I reports three metrics per method and scenario: energy of the most loaded drone (E_o), total traveled distance (l) and computation time (t_c). For the stochastic solvers, results are reported as the mean $\pm$ standard deviation over five independent runs with different random seeds, providing an indication of both average performance and variability across runs. DARP is deterministic and is therefore evaluated using a single run per scenario, configured with a coverage density equivalent to that of the other methods so that the energy comparison stays fair.

Energy-ALNS lowers E_o in all six near-convex scenarios (Rectangle and Simple), with gains between 2.9% and 8.6% over MSTSP. In these six cases Energy-ALNS also returns the same solution across the five seeds, while MSTSP varies by up to 3.5 Wh. In the remaining ten scenarios, the differences between MSTSP and Energy-ALNS are considerably smaller, generally remaining below 0.5%. Given the stochastic nature of both methods, these small differences should be interpreted together with the variability observed across independent runs, which reaches 1.5 Wh for MSTSP in Complex with 3 UAVs. Two patterns deserve attention. First, DARP, evaluated at the same coverage density as the other methods, spends more energy than MSTSP and Energy-ALNS in all 16 cases. Its geometric partition, efficient in distance within its original paradigm, loses competitiveness once the same energy model and the same coverage resolution score it. Second, DR-ALNS records $t_c \approx 0$ s in Rectangle and Simple because these areas end up with about one cell per drone: BCD yields few cells there, and the reference pipeline splits the largest cells only until each UAV can receive one [1]. In this trivial case the PPO policy runs no real search iteration.

C. Aggregate Performance

Fig. 4 plots the percentage gain of Energy-ALNS over MSTSP for each of the 16 scenarios.

Energy-ALNS cuts the energy makespan in 11 of the 16 scenarios, averaging $+1.90\%$ over MSTSP. The largest gains land in near-convex scenarios: Simple with 4 UAVs reaches $+8.6\%$ and Rectangle with 2 UAVs reaches $+3.4\%$. Island ties in every case and Complex with 2 UAVs records -0.1%.

D. Operator Study

We kept DR-ALNS in the comparison to test whether a reinforcement learning controller would add anything to Energy-ALNS. In its original form, with distance-oriented operators, DR-ALNS trails the other solvers in Table I. Two staged interventions isolate the source of the energy gain. In the first, we swapped the generic operators for the energy-aware operators we designed for Energy-ALNS and left the DR-ALNS selection policy untouched. The mean gain against MSTSP moved from -39.15% to -1.36%, and the balance index across routes fell from 1.37 to 1.02. In the second, we retrained the selection policy with a dense reward on top of the energy-aware operator set. The solutions matched the previous configuration in 15 of the 16 scenarios, and even with the energy-aware operators DR-ALNS never overtook Energy-ALNS. Both interventions point to the operator set as the component that carries the gain, while the learned controller adds no measurable improvement once the operators fit the energy objective.

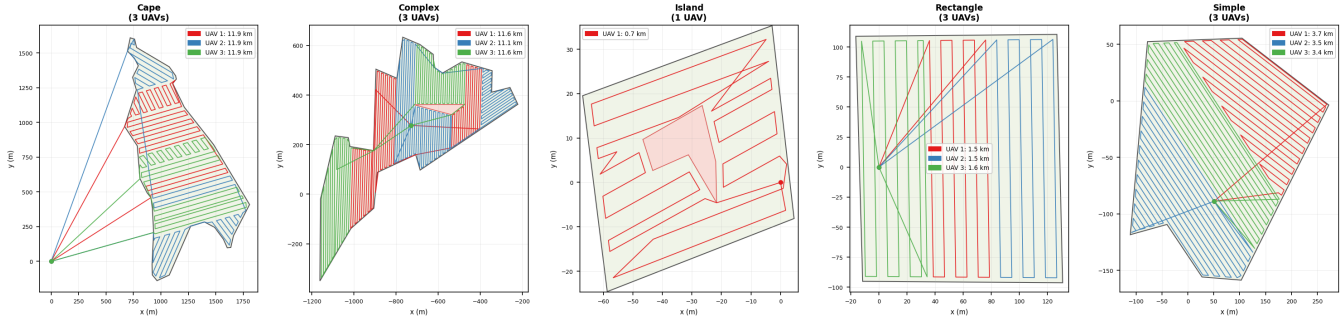


Fig. 3. Representative coverage routes generated by Energy-ALNS for the five scenario geometries: Cape (non-convex with indentations), Complex (irregular non-convex), Island (central no-fly zone), Rectangle (convex) and Simple (one reflex vertex). Each color marks the route of one UAV, and the legend reports the distance flown by each vehicle.

TABLE I

COMPARISON ACROSS METHODS: ESTIMATED ENERGY E_o [WH], TOTAL DISTANCE l [KM] AND COMPUTATION TIME t_c [S]. STOCHASTIC SOLVERS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE INDEPENDENT RANDOM SEEDS. DARP IS DETERMINISTIC (SINGLE RUN). BOLD MARKS THE LOWEST E_o OF EACH ROW.

Group	Scenario	MSTSP			Energy-ALNS			DARP			DR-ALNS		
		E_o [Wh]	l [km]	t_c [s]	E_o [Wh]	l [km]	t_c [s]	E_o [Wh]	l [km]	t_c [s]	E_o [Wh]	l [km]	t_c [s]
Cape	2 UAV	**268.8** ± 0.0	32.6	1.2	**268.8** ± 0.0	32.6	2.4	418.4	47.5	127.6	510.0 ± 0.0	34.3	1.4
	3 UAV	202.8 ± 0.9	36.1	5.0	**202.2** ± 0.2	36.0	6.6	288.7	47.4	124.9	281.8 ± 0.0	37.3	2.4
	4 UAV	168.0 ± 1.0	40.0	13.4	**167.2** ± 0.4	39.4	12.2	214.8	47.5	111.1	406.3 ± 6.5	41.8	2.7
Complex	2 UAV	**291.7** ± 0.4	33.2	1.3	291.9 ± 0.0	33.2	4.0	563.8	57.9	429.6	374.6 ± 0.0	33.9	1.4
	3 UAV	201.7 ± 1.5	34.4	3.2	**201.2** ± 0.6	34.3	5.6	388.4	57.9	238.7	300.1 ± 0.0	35.3	1.9
	4 UAV	155.8 ± 0.9	35.1	8.4	**155.4** ± 0.9	35.0	10.1	303.0	58.0	351.6	279.4 ± 0.0	35.9	2.9
	Large (3 UAV)	323.5 ± 0.1	56.0	2.8	**323.4** ± 0.0	57.1	6.1	620.6	98.8	385.8	494.0 ± 18.4	57.4	1.9
Island	1 UAV	**18.9** ± 0.0	0.7	0.1	**18.9** ± 0.0	0.7	0.6	36.5	1.3	7.2	22.2 ± 0.0	0.8	1.1
	2 UAV	**10.7** ± 0.0	0.7	0.1	**10.7** ± 0.0	0.7	0.4	23.7	1.3	8.4	16.1 ± 1.1	0.8	1.1
	3 UAV	**7.9** ± 0.0	0.8	0.2	**7.9** ± 0.0	0.8	0.6	16.5	1.2	18.5	16.3 ± 0.9	0.9	1.2
Rectangle	2 UAV	40.3 ± 0.9	4.2	0.0	**38.9** ± 0.0	4.2	0.1	77.0	7.4	31.3	39.4 ± 0.0	4.2	0.0
	3 UAV	28.5 ± 0.7	4.4	0.0	**27.7** ± 0.0	4.4	0.1	59.4	7.4	10.5	28.0 ± 0.0	4.5	0.0
	4 UAV	23.2 ± 0.4	4.7	0.0	**22.5** ± 0.0	4.5	0.2	46.9	7.4	10.4	23.2 ± 0.0	4.8	0.0
Simple	2 UAV	93.2 ± 2.6	10.1	0.0	**89.7** ± 0.0	10.0	0.2	183.5	18.2	102.8	90.8 ± 0.0	10.1	0.0
	3 UAV	68.0 ± 1.7	10.4	0.0	**63.0** ± 0.0	10.3	0.3	136.5	18.1	42.1	63.0 ± 0.0	10.3	0.0
	4 UAV	54.7 ± 3.5	11.0	0.0	**50.0** ± 0.0	10.6	0.4	104.3	18.1	39.7	50.2 ± 0.0	10.6	0.0

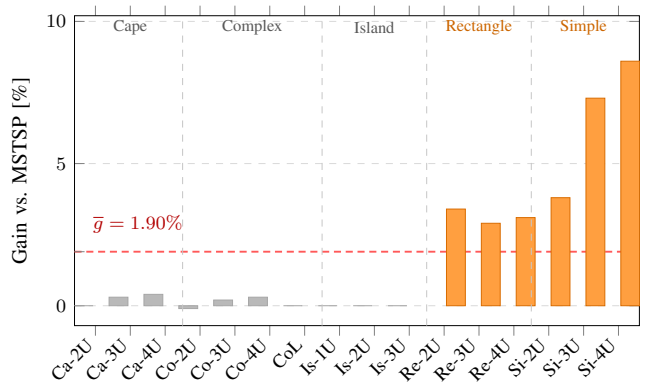


Fig. 4. Percentage gain of Energy-ALNS over MSTSP across the 16 scenarios (mean over 5 seeds). Orange bars mark near-convex scenarios and gray bars mark irregular scenarios or scenarios with a no-fly zone. The dashed red line marks the mean gain of $+1.90\%$. Ca = Cape, Co = Complex, CoL = Complex Large, Is = Island, Re = Rectangle, Si = Simple; the number after the hyphen gives the fleet size.

E. Discussion

The results split into two regimes. In near-convex scenarios with no obstacle, the search space allows free redistribution of cells across routes, and Energy-ALNS turns that freedom into a consistent gain over MSTSP. In irregular scenarios or scenarios with an exclusion zone, geometry already blocks most of the profitable exchanges, so both solvers converge to close or identical solutions.

DARP offers a useful contrast because it partitions the area by a geometric criterion anchored on the initial positions of the UAVs instead of solving the MS-TSP formulation. Scored at the same coverage density, its energy consumption nearly doubles that of Energy-ALNS in several cases, such as 288.7 Wh against 202.2 Wh in Cape with 3 UAVs. We keep it in Table I as a reference for that alternative paradigm and leave it out of Fig. 4.

The operator study confirms that the choice of destroy and repair operators weighs more than the mechanism that selects them. Energy-ALNS, a classical ALNS carrying operators matched to the energy objective, beats an ALNS with reinforcement learning whose operators were designed for a different one.

F. Simulation Results

We exported the routes produced by Energy-ALNS to a simulation environment built from ROS 2 Humble, Gazebo Sim 8 and the PX4 stack. Gazebo reproduces multirotor dynamics with mass, actuator and attitude control models. The test checks executability, that is, whether an offline plan becomes a flyable trajectory, and does not measure energy. We chose Island with 3 UAVs because its central no-fly zone exercises the exclusion constraints under vehicle dynamics.

The three UAVs completed the coverage mission, respected the exclusion zones and returned to their home positions with no manual intervention, and the PX4 stack read the exported routes as written. Energy-ALNS and MSTSP tie in Island (Table I), so this run does not show that the predicted energy savings survive real dynamics; measuring flown energy in a near-convex case such as Simple with 4 UAVs remains open.

V. Conclusion

We replaced the MSTSP solver of an energy-aware multi-UAV coverage planner with Energy-ALNS, an Adaptive Large Neighborhood Search specialized for battery consumption. It keeps the Boustrophedon decomposition, the energy estimation and the MS-TSP construction of the reference method, changing only how the planner selects, assigns and orders the patterns across the fleet. The solver combines a GRASP initial solution, seven energy-guided removal operators, greedy and regret-2 repair, an intra-route local search coupled with pattern orientation optimization, and a final inter-route energy rebalancing stage, all scored by the same cost function.

Over 16 scenarios with five seeds per solver, Energy-ALNS cut the energy makespan of MSTSP by 2.9–8.6% in the six near-convex scenarios, while the other ten stayed within seed variance. The operator study traced this gain to its source: swapping generic operators for energy-aware ones inside DR-ALNS moved its result from -39.15% to -1.36% against MSTSP, while retraining the selection policy left 15 of the 16 solutions unchanged. Operators specialized in energy, not the reinforcement learning component, decide how competitive the solver is.

Two limitations bound these conclusions. DARP and DR-ALNS start from formulations other than the shared MS-TSP one, which narrows the comparison in scenarios with different fleet sizes, and validation stayed inside simulation due to the lack of physical drones.

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